

Editorial

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Welcome to the Summer 2026 issue of the International Journal of Microsimulation. The five papers collected here range from early-childhood policy and social-security financing to food carbon pricing, social assistance and urban traffic. Their subject matter is unusually broad, but they share a concern with balancing the approximations built into the microsimulation models and discussing this choice explicitly. Each asks, in a different way, what must be represented in detail, what can be simplified, and how the resulting model can be made credible for the decision at hand. The issue therefore illustrates both the breadth of contemporary microsimulation and a common methodological theme: fit-for-purpose modelling.

The issue opens with Venkatesh and colleagues, "LifeSim Childhood: Extrapolating Intervention Effects and Public Cost Savings from Birth to Adolescence in the UK". LifeSim Childhood is a general-purpose model designed to translate changes in circumstances during childhood into a broad set of later health, educational, social, public-cost and wellbeing outcomes up to age 17. Its distinctive methodological choice is a regression-based 'long-jump' approach: rather than modelling every intervening transition in a relay from one age to the next, it estimates total effects of exposures on later outcomes directly from the Millennium Cohort Study, with confounder selection guided by explicit causal assumptions. The paper illustrates the model using hypothetical increases in early-childhood income and compares the resulting effect magnitudes with external experimental and quasi-experimental evidence. The value for readers is methodological as much as substantive: the paper provides a transparent way of bringing long-term and cross-sectoral consequences into the appraisal of childhood interventions when direct long-term experimental evidence is unavailable. In the income illustration, the economic case is driven more by estimated wellbeing gains than by savings to public services, a useful reminder that fiscal savings capture only one part of the returns to early intervention.

Liégeois, in "Alternatives for financing social security in Luxembourg by resident and cross-border households", turns to a very different policy problem: how to finance a social-security system under demographic pressure in an economy where cross-border workers are central to both employment and public revenues. The analysis uses two EUROMOD-type models based on Household Finance and Consumption Survey data, one for resident households and one for cross-border households, and compares 42 parametric alternatives affecting social contributions and personal income taxation. The alternatives are assessed against three deliberately simple dimensions: total receipts, poverty and inequality, with a basic composite performance index used to help navigate the resulting trade-offs. The paper is useful precisely because it does not pretend that the exercise identifies a uniquely 'best' reform. Different instruments move revenues and distributional outcomes in different directions, and the preferred option depends on social preferences and political feasibility. The broader lesson is that a transparent static, 'day-after' simulation can be an appropriate first stage of policy design, before adding labour-supply responses, partial-equilibrium effects or longer-term dynamics.

Al-Masbhi and colleagues widen the scope again in "A Comparative Methodological Framework for Evaluating Distributional and Nutritional Effects of Food Carbon Pricing". The paper links Life Cycle Assessment emission intensities with Household Budget Survey microdata, retail prices and nutritional information for seven EU Member States and Turkey. A harmonised static microsimulation applies a carbon price to the emissions embodied in food purchases under full pass-through and fixed quantities, allowing a comparable assessment of food expenditure, diet-related emissions, the cost of dietary energy and tax incidence across the income distribution. The results point to small but consistent regressive effects across countries, while the sensitivity analysis suggests that alternative

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LCA emission factors do not overturn the qualitative distributional patterns. The framework is best read as a first-round incidence benchmark rather than a complete model of behavioural adjustment. Its practical contribution is to make environmental, nutritional and distributional information speak to each other within one household-level framework, while also showing clearly where the next modelling steps lie: substitution responses, alternative tax designs and revenue recycling.

Tervola's "Refining the simulation of social assistance with monthly income data and calibration" focuses on a narrower modelling choice with surprisingly large consequences. In the Finnish SISU tax-benefit model, social assistance had traditionally been simulated largely from annual income, even though eligibility is determined month by month. The paper combines estimated monthly incomes with a regression-based calibration procedure aimed at correcting oversimulation arising from non-take-up and other unobserved eligibility determinants. The refinements reduce the overall recipient misclassification rate and the error in simulated benefit amounts. More importantly, they materially change the estimated consequences of the 2024–2025 benefit reforms: the increase in social-assistance recipient households falls from about 50,000 to 26,000, while the estimated increase in child poverty is around 60% larger. The take-home message extends well beyond Finland. Matching aggregate expenditure or recipient totals is not sufficient validation when policy conclusions depend on who receives a benefit and for how many months; time resolution and micro-level calibration can change the distributional story.

The final contribution, "Methodological Foundations of the Traffic Simulation Process", takes the journal into urban mobility. It offers a software-independent practical framework covering microscopic, mesoscopic and macroscopic traffic simulation, from the definition of objectives and spatial scale through data collection, travel-demand estimation and model construction to calibration, validation and scenario analysis. The paper also reviews common traffic-behaviour models and compares widely used open-source and commercial platforms, emphasising that the appropriate tool depends on data, computational resources and the decision problem rather than on a universal ranking of software. For readers whose main experience is with socio-economic microsimulation, the application domain is different but many of the methodological questions are familiar: how much granularity is useful, which inputs need to be observed, what should be calibrated, and which outputs constitute convincing validation. The paper is therefore both a practical guide to traffic microsimulation and a useful reminder of the wider family of unit-level simulation methods to which our field belongs.

Taken together, the five papers also make a useful point about model complexity. More detail is not automatically more informative. In the Finnish application, greater temporal granularity improves the simulation of a monthly benefit; LifeSim Childhood deliberately avoids modelling every mediating pathway; the Luxembourg analysis deliberately starts from a static framework; the food-carbon application fixes quantities to obtain a transparent first-round benchmark; and the traffic paper treats modelling scale as a choice that should follow the objective and the available data. Across very different applications, credibility comes less from maximising realism in every dimension than from making simplifying choices visible, testing their consequences where possible, and matching model architecture to the policy question.

Suggestions for further reading

A direct companion to the LifeSim Childhood paper is *Venkatesh et al. (2026)*, which applies birth-cohort microsimulation to compare the social costs of six early-years disadvantages in the UK and Bradford. The paper usefully shows that priorities can differ depending on whether costs are considered per affected child or for the population as a whole.

For readers interested in extending first-round carbon-pricing incidence analysis, *Kettner et al. (2026)* link a macroeconomic model with household microsimulation to study carbon pricing and alternative revenue-recycling schemes in Austria. The analysis illustrates how macroeconomic feedbacks, regional heterogeneity and compensation design can alter the distributional assessment of carbon pricing.

Also related to the content of this issue, *Figari et al. (2026)* develop a spatial extension of EUROMOD to estimate the local anti-poverty effects of child benefits in Italy and Greece. Their results reveal substantial within-country heterogeneity, illustrating the information that may be hidden by national averages in otherwise standard tax-benefit microsimulation.

Finally, Associate Editor Peng Zhang recommends “Visions of Inequality: From the French Revolution to the End of the Cold War” by Branko Milanovic (**Milanovic, 2023**). This book offers a profound historical perspective on economic inequality, tracing the intellectual evolution of the inequality debate over two centuries and proposing a ‘gold standard’ for research that seamlessly integrates narrative, theory, and empirical evidence. For our community, its greatest value lies in highlighting the historical roots of micro-level analysis. By bridging the foundational economic theories with empirical distributional data, Milanovic provides a framework for understanding how micro-level dynamics shape macro-level inequality. This book serves as an excellent reminder of the theoretical underpinnings of our technical methods and encourages researchers to ground their microsimulation models in robust, historically informed economic narratives.

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